CS 540 — Linear Regression Exercises

More Than One Input & Vector Equations

Solutions to these exercises are available in `6_lin_reg.org` on GitHub. Work through the questions yourself first — these are meant to build intuition for reading loss surfaces and writing model equations, not to be looked up directly.

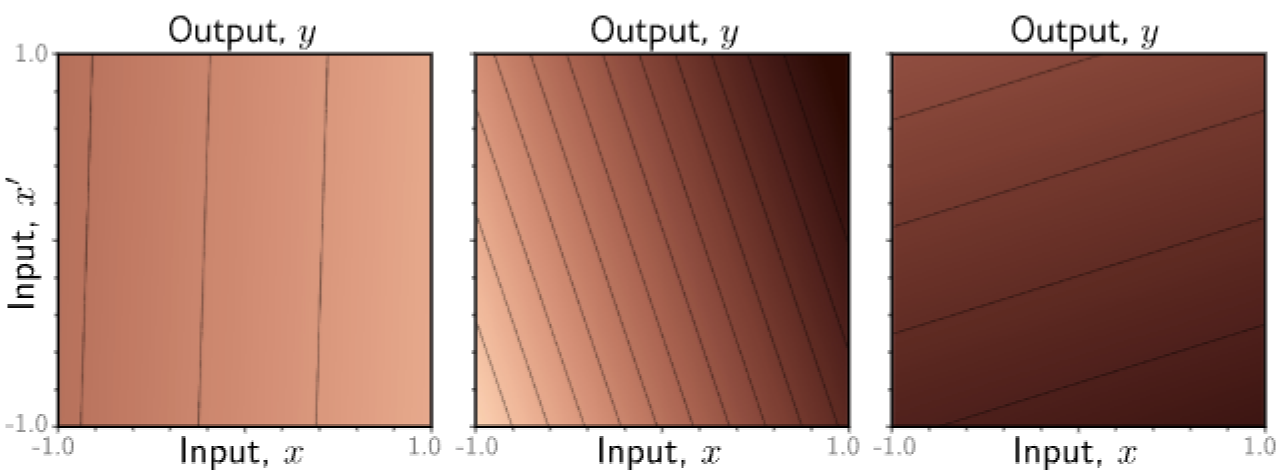
Exercises: More Than One Input

How to read one of these heatmaps:

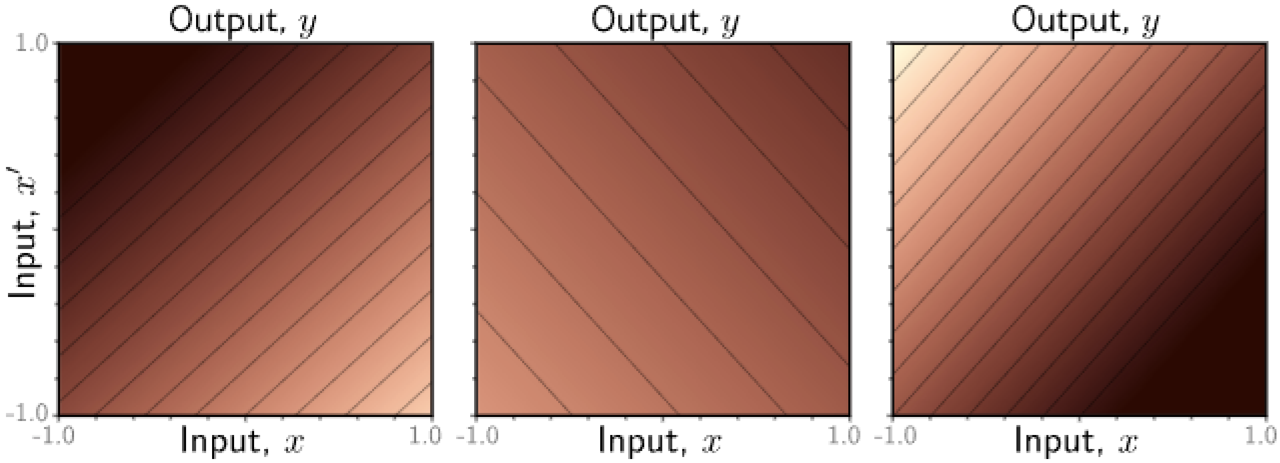
- φ_0 only sets the overall brightness (a uniform shift) - by itself it creates no pattern at all. If $\varphi_1 = \varphi_2 = 0$, the heatmap is a single flat color with no isolines whatsoever.
- φ_1 controls the left-right (x_1) trend, φ_2 controls the top-bottom (x_2) trend. Read each one's **sign** off the direction of the change - e.g. does it get brighter or darker as x_1 increases - not off which corner happens to look dark, since both parameters' effects combine in the corners.
- The vector (φ_1, φ_2) is the **gradient**: it points toward the steepest increase (the brightest direction), and its magnitude \sqrt{\phi_1^2+\phi_2^2} sets how fast the color changes.
- The **isolines** (contour lines) are always perpendicular to the gradient, with slope $-\varphi_1/\varphi_2$. Closely packed isolines mean a large gradient magnitude (steep, fast-changing); widely spaced isolines mean a small one (shallow, slow-changing).

Use these insights to answer the following questions:

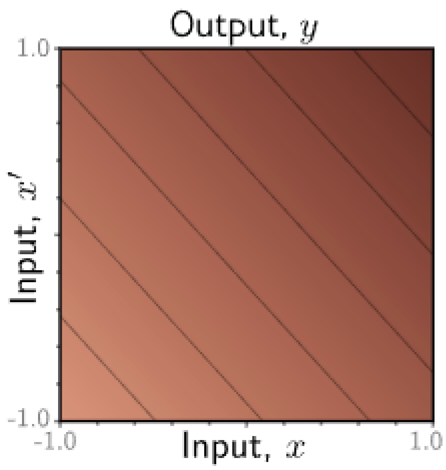
1. What do you expect the heatmap representing the model to look like when the parameters φ are $\varphi_0 = 1.0$, $\varphi_1 = \varphi_2 = 0$?
2. What do you expect the heatmap to look like for $\varphi_0 = 1.0$, $\varphi_1 = 0$, $\varphi_2 = -0.5$?
3. Which of these 2D linear models has parameters $\varphi_0 = -0.33$, $\varphi_1 = -0.11$, $\varphi_2 = 0.35$?



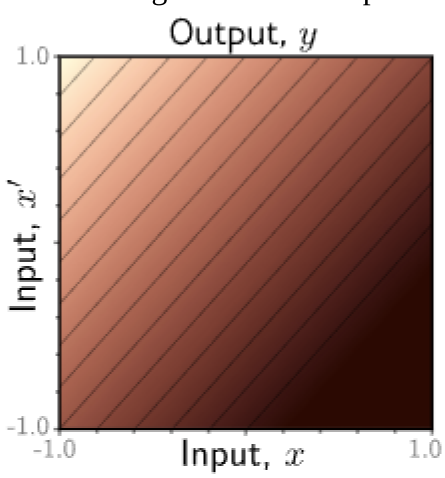
4. Which of these 2D linear models has parameters $\phi_0 = -0.0$, $\phi_1 = 0.75$, $\phi_2 = -0.82$?



5. Guess the rough values of the parameters for the model shown here:



6. Guess the rough values of the parameters for the model shown here:



Exercises: Vector Equations

How to build one of these vector-form equations:

- The **vector form** is still just a dot product, as in the plain linear model: stack every term's coefficient into ϕ and every matching power/product of the inputs into a second vector, then $f[x,\phi] = \phi^T \cdot$ [\text{terms}]. Going to higher order or more inputs doesn't change this structure - it only makes the two vectors longer.

- To list the terms for order m, include **every** monomial in the inputs with total degree from 0 up to m - don't skip any. Leaving one out isn't just a bookkeeping shortcut: each term is what lets the model bend a specific way, so dropping it removes a real shape the model could otherwise fit (e.g. dropping x^2 stops it being quadratic at all).
- With **more than one input**, distinguish pure-power terms (x^2, x'^2) from **interaction** terms (x x'). Without the interaction term(s), the model is **separable** - just a sum of independent curves, one per input - which restricts it to an axis-aligned bowl or saddle. Interaction terms are what let the isolines tilt off-axis, the same role the ratio φ_1{:}φ_2 played for the plain linear model.
- The parameter **count** for order m with D_i inputs is the number of monomials of total degree ≤ m in D_i variables - a binomial coefficient (choosing with replacement), not a simple product like $m \times D_i$.
- **Multiple outputs** need no new equation form: stack independent copies of the same model, one full φ vector per output, all sharing the same input vector.

Use these insights to answer the following questions:

1. What is the equation for a quadratic model with one input x and one output in vector form?
2. What is the equation for a cubic model with a single input x in vector form?
3. What is the equation for a quadratic model with two inputs x and x'? Why?
4. Consider a polynomial model of order m with D_i inputs and one output. How many parameters will this model have? Give the answer in terms of m and D_i.

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